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Repeated-measures treatment comparison -- Statistical Report

Template: nature | **Generated**: 2026-10-19 20:06 中国标准时间 | **Pipeline**: data-master
Raw input: `measurements.csv` (sha256 `2519c6d2304dc381...`) | **Analysed file**: `cleaned.csv`
Rows: 36 raw -> 36 analysed | **Columns**: 4 | **Modality**: tabular
Independent unit: every unit belongs to exactly one group; multiple rows per unit are replicates, and the primary analysis is run on per-unit means
Types: `cleaning_run.json` (4 columns replayed from the cleaning manifest, never re-guessed)

Executive summary

Setup. 36 rows and 4 columns reached the analysis, of which 2 are numeric measurements, compared across 2 levels of `arm` (control, treatment).

The conflict. No defect capable of changing a conclusion was detected in the raw file.

How it was resolved. 9 documented operation(s) were applied, 36 rows became 36, and the cleaned file reproduces byte for byte. The clearest separation is that `response` is compared across arm: Cohen's d = -6.35, 95% CI -9.36 to -3.35, BH-adjusted p = 6.6e-07 across a family of 1 outcomes.

The caveat that matters most. No defect requiring a caveat was detected in this run. That is a statement about the checks that ran, not a guarantee that the data are fit for the conclusions drawn.

1. Setup

- **Source**: `measurements.csv`, read in `text` mode so that no column type was guessed at load time -- a reader that guesses turns a postcode `02134` into `2134` before any audit of the decision can exist.
- **Routing**: modality `tabular`; 4 columns profiled.
- **Grouping variable**: `arm`, which the router scored highest among the columns able to form groups (score 1.000, balance 1.00 as entropy normalised by log of the level count). Grouping by a column with a single level would have returned one group compared with itself and a confident, meaningless zero.
- **Row meaning**: the router suggests `row`; that is a heuristic, and the real answer is an open question below.
- **Pre-registration**: AUTHOR_INPUT_NEEDED. Nothing in the input says whether these analyses were planned in advance, and the outcome columns were chosen after the data were seen.

2. Methods

2.1 Cleaning

The raw file was never modified. 9 operation(s) were applied automatically and 0 were withheld for a human decision. Every applied operation names the columns it is permitted to touch; the run then compares raw against cleaned in text space and fails if any column no operation named gained a value. In this run: all 2 untouched columns kept exactly their raw values.

Byte-level reproducibility: passed -- hash matches. Cleaned sha256 `6f62b85a4cf5d537...`.

> high-risk operations are never silent; see references/never-automatic-list.md

2.2 Types

Types were decided once and replayed, never re-guessed: 4 columns were typed from the manifest in `cleaning_run.json`. A column that the cleaning stage declared textual cannot quietly become numeric in the analysis stage while the report still describes it as an identifier. Dtype source for the analysis stage: `cleaning_run.json`.

2.3 Missing data

No variable carries missing values.

2.4 Test selection

Outcome	Design	Method	Statistic	df	p (raw)	p (corrected)	Alternatives rejected
response	2-group numeric	Student two-sample t-test (pooled variance)	t = -10.999	10	= 6.6e-07	= 6.6e-07	Welch t-test: equal variance holds (Levene p > alpha); the pooled test is more powerful

2.5 Multiple comparisons

The primary family holds **1 outcomes**, one per measured variable, corrected by **BH** at alpha = 0.050. 1 of 1 survive the correction. one outcome = one member of the family; correction applies across outcomes, and separately within each post-hoc pairwise family

3. Results

3.1 What is in the table

Column	Role after cleaning	n	% missing	Distinct values	Status
subject	categorical	36	0.0	12	kept
arm	categorical	36	0.0	2	kept
visit	numeric	36	0.0	continuous	kept
response	numeric	36	0.0	continuous	kept

3.2 Descriptive statistics

Variable	n	% miss	Mean (95% CI)	SD	Median (IQR)	Range	Skew
visit	36	0.0	2.00 (1.73 to 2.27)	0.83	2.00 (1.00-3.00)	[1.00, 3.00]	0.00
response	36	0.0	10.71 (10.44 to 10.97)	0.80	10.40 (10.10-11.43)	[9.70, 12.30]	0.73

Mean and median are both given because they disagree wherever the distribution is skewed. The mean column carries a normal-approximation interval, which for a skewed variable is the least trustworthy number on this table; the median and IQR beside it are not.

3.3 Categorical variables

Variable	n	Distinct	Mode	Distribution
subject	36	12	c01	c01 (3), c02 (3), c03 (3), c04 (3), c05 (3), c06 (3)
arm	36	2	control	control (18), treatment (18)

3.4 Group comparison

Outcome	Method	Statistic	n	Effect size (95% CI)	p (raw)	p (BH-adjusted)
response	Student two-sample t-test (pooled variance)	t = -11.00	12	Cohen's d = -6.351 (95% CI -9.36 to -3.35)	= 6.6e-07	= 6.6e-07

Read the effect-size column before the p column. response is compared across arm: Cohen's d = -6.35, 95% CI -9.36 to -3.35, BH-adjusted p = 6.6e-07 across a family of 1 outcomes. The ordering by effect size and the

ordering by p value are not the same statement, and at this sample size the p values are small enough that their exact magnitude carries no information beyond "smaller than this table can resolve".

4. Discussion

What the data supports. The measurements differ across levels of `arm` by margins larger than the sampling noise at this size: `response` is compared across arm: Cohen's d = -6.35, 95% CI -9.36 to -3.35, BH-adjusted p = 6.6e-07 across a family of 1 outcomes.

What the data does not support. Nothing here establishes that the group label produced the measurement. The design compares existing groups rather than assigning them, and species, batch, site, instrument and operator all travel together with the group label -- this table cannot separate them. A claim of the form "X caused Y" is therefore not available from these data, and none is made in this report.

Boundary conditions. These results describe the rows in this file, at these measurement conditions, with this instrument. The intervals quantify sampling variation inside that set; they say nothing about another lot, another season or another population. The corrected family covers the outcomes tested here, so an outcome that was not tested is not thereby null.

5. Caveats and limitations

C1. no defect detected. No defect requiring a caveat was detected in this run. That is a statement about the checks that ran, not a guarantee that the data are fit for the conclusions drawn.

- *Limits*: all claims
- *Status*: not applicable

6. Reproducibility

Every number above is traceable to a script. Nothing in this report was typed by hand.

Stage	Script	Artifact	sha256 (first 16)
Profile	scripts/profile_tabular.py	analysis/profile.json	cd348014d1cca169
Plan	scripts/cleaning_plan.py	cleaning_plan.json	7ac8130efd147906
Clean	scripts/execute_cleaning.py	cleaned/cleaned.csv	6f62b85a4cf5d537
Validate	scripts/validate_clean.py	cleaned/validation.json	ad2137c384c90c6e
Describe	scripts/stats_descriptive.py	analysis/descriptive.json	6cbb044156c80a75
Infer	scripts/stats_inferential.py	analysis/inferential.json	777cf97debe33ea1
Decide figures	scripts/chart_decision.py	analysis/chart_decisions.json	a6d3103be5b99c94
Render figures	scripts/render_chart.py	charts/*.png	e1a6abf058364938
Assemble	scripts/assemble_report.py	report.md / report.pdf	self-referential -- see report_build.json

A document cannot contain its own hash, so the hash of `report.md` and `report.pdf` is recorded in `report_build.json` rather than on this page. Every other hash above was computed from the artifact as it exists now.

- **Interpreter**: Python 3.13.15 on Windows-11-10.0.26200-SP0
- **Libraries**: numpy 2.4.6, pandas 3.0.3, scipy 1.18.1
- **Pipeline code**: data-master commit `8749c5ea1ff4` (working tree was dirty; the commit alone does not reproduce it) -- this report names the exact code that produced every number in it.
- **Random seeds**: 42 -- every resampling interval is seeded, so it reproduces exactly.
- **Columns any operation was permitted to touch**: `response`, `visit`

- **PDF backend**: `reportlab`. The page count, byte count and sha256 of the delivered PDF are recorded in `report_build.json` under `pdf`; the probe that established this backend is under `pdf_probe`. They are not printed here because a document cannot contain its own hash, and a page count measured before the last line changed is not the page count of the file you are reading.

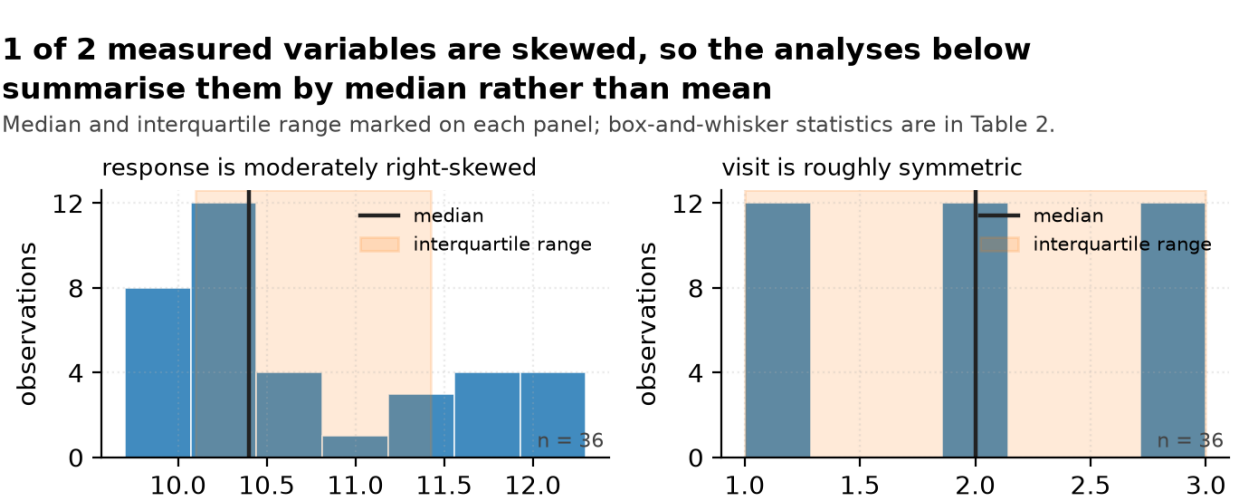
The data contains non-Latin characters, so the renderer was asked for a CJK-capable font; one was embedded.

7. Figures

Each figure carries one argument and its title states that argument as a finding computed from the data rather than as a topic label. The chart types considered and rejected are listed under each one, because choosing a chart is itself a claim about what matters.

Figure 1 -- 1 of 2 measured variables are skewed, so the analyses below summarise them by median rather than mean

Distribution of each numeric variable, with the median and interquartile range marked. Panel order follows absolute skewness. Full box-and-whisker statistics are in Table 2.



- **What it shows**: response is moderately right-skewed (median 10.40, IQR 10.10 to 11.43, n = 36); visit is roughly symmetric (median 2.00, IQR 1.00 to 3.00, n = 36).
- **How to read it**: Skewed variables are summarised by the median and IQR, and rank-based or Welch methods are preferred for them.
- **What follows**: These summaries, not the raw shape, should carry into any downstream model.
- **Uncertainty shown**: median and interquartile range marked on every panel
- **Chart alternatives rejected**: one histogram per variable as separate figures -- same argument repeated; the reader has to re-read the framing each time; box plot only -- understates the tail, which is precisely what decides the test choice
- **Source**: `scripts/chart_decision.py` -> `charts/fig01_distribution_panel.png` rendered by `scripts/render_chart.py`, 89,713 bytes, ink ratio 0.4227

8. Figure specifications

#	ID	Chart type	Specification
1	`fig01_distribution_panel`	distribution_panel	2 panels; bins = auto; show_median_iqr = True

Appendix A1. Sensitivity and stability

The primary analysis was re-run under perturbation by `scripts/sensitivity_analysis.py` (seed 42, 10 blocks for the jackknife). Agreement means the significance verdict did not flip at alpha=0.05.

Outcome	Perturbation	Statistic	p	Verdict
`response`	strict_complete_case (n=36 of 36 rows)	Student two-sample t-test (pooled variance) = -11.00	= 6.6e-07	agrees
`response`	row_level_comparison	Welch two-sample t-test (unequal variance) = -6.97	= 8.3e-07	agrees
`response`	rank_cross_check	Mann-Whitney U = 0.00	= 0.002	agrees

Assumption robustness (distinct from the perturbations above):

Outcome	Check	Statistic	p	Verdict
`response`	rank-based cross-check of the primary test	Mann-Whitney U = 0.00	= 0.002	agrees

Each primary parametric test was repeated with its rank-based equivalent. Agreement means the conclusion does not depend on the normality assumption the parametric test makes; a disagreement would have meant the parametric test had to go, not be defended.

A1.1 Sensitivity analyses not run

None -- every perturbation that applies to this design ran, and the ones that do not apply are marked as such in `analysis/sensitivity.json`.

Appendix A2. Cleaning ledger

Every operation that ran, in order, with the catalogue rule it came from. Nothing below was chosen by the model at run time: the plan is the gate, and only `plan["auto_actions"]` may execute.

#	Operation	Columns	Sev.	Rows	Rule
1	`fix_unicode`	`arm`, `response`, `subject` (+1 more)	P3	36->36	`t-fix-unicode`
2	`strip_whitespace`	`arm`, `response`, `subject` (+1 more)	P3	36->36	`t-strip-ws`
3	`collapse_whitespace`	`arm`, `response`, `subject` (+1 more)	P3	36->36	`t-collapse-whitespace`
4	`trim_paired_quotes`	`arm`, `response`, `subject` (+1 more)	P3	36->36	`t-trim-quotes`
5	`coerce_numeric`	`response`, `visit`	P2	36->36	`t-coerce-numeric`
6	`normalize_dtypes`	`arm`, `response`, `subject` (+1 more)	P2	36->36	`t-normalize-dtypes`
7	`drop_all_null_cols`	(file-level)	P1	36->36	`t-drop-all-null-cols`
8	`drop_all_null_rows`	(file-level)	P1	36->36	`t-drop-all-null-rows`
9	`rename_columns_append_cleaned`	(file-level)	P2	36->36	`t-rename-clean`

The `Rule` column names the catalogue anchor; the full path, the operation id and the exact column list for every row above are in `cleaned/cleaning_run.json` under `applied_ops`.

Withheld -- proposed but deliberately not applied.

Nothing was withheld.

Failed -- attempted and did not complete.

No operation failed.

Additive-only audit. The contract forbids modifying the raw file and permits an operation to touch only the columns it names. Columns no operation named: **2, and each kept exactly its raw values -- audit passed**.

Appendix A3. Alternatives considered and not used

A test is a choice. Listing only the test that was run hides the fact that a different, defensible analysis was available, so every rejected alternative is recorded here with the reason it lost.

Outcome	Design	Rejected test	Reason
`response`	2-group numeric	`Welch t-test`	equal variance holds (Levene p > alpha); the pooled test is more powerful

The rule that made these choices is `references/nature-statistics-checklist.md#gates-1-3-5`. It was applied mechanically; there is no per-dataset override, because a rule that is quietly overridden is a rule that cannot be checked.

Appendix A4. Where every number came from

Each value below was registered by the code that read it from an artifact on disk. A sentence in this report that is not backed by a row here is a sentence that should not have been written.

Kind	Value	What it means	Source: script -> artifact	Rule anchor
`header`	36	rows analysed	`execute_cleaning.py` -> `cleaning_run.json`	`--`
`headline`	`response` is compared across arm: Cohen's d = -6.35, 95% CI...	strongest group separation	`stats_inferential.py` -> `inferential.json`	`--`
`rejected test`	Welch t-test	rejected for response	`stats_inferential.py` -> `inferential.json`	`two_group_numeric`
`correction`	BH	family correction method	`stats_inferential.py` -> `inferential.json`	`gate-5`

Only the basename of each artifact and script is shown here; the full path, the untruncated value and the complete claim record for every row are in `report_build.json` under `provenance`. Each record carries `rule_source`, and where a rule had an alternative the record names it and why it lost -- a claim with no alternative says so rather than leaving the field blank.

Appendix B. Open questions the data cannot answer

1 question must be answered by someone who knows how the data were collected. They are listed with the consequence of leaving them unanswered, because an unanswered question that carries no stated cost is one nobody will prioritise.

#	Question	Why it matters	Cost of leaving it open
1	Were these outcomes chosen in advance?	Nothing in the input records a pre-specified analysis plan, and the outcome family was assembled from the columns present. Without that, the family-size correction controls multiplicity within this report but cannot address selection across reports.	HIGH -- determines whether the correction is the right one

What this report does not claim

- It does not claim the group label produced the measurement.
- It does not claim the analysis was pre-specified.
- It does not claim the rows are independent observations.
- It does not claim any mechanism for the missingness it reports; the screening test in 2.3 is a description, not a diagnosis.

End of report. Generated 2026-10-19 20:06 中国标准时间 *by data-master. The raw input was never modified: every value above was read from the cleaned file or from a recorded artifact. Each number is registered in `report_build.json`; the gate verdict, including anything that failed, is in `audit/final_audit.md`.*